

# SpineMRI Foraminal Stenosis Grader

Clinical AI Report · Generated 14 July 2026, 11:15

|            |                  |          |                  |
|------------|------------------|----------|------------------|
| Patient ID | Not specified    | Model    | ConvNeXt V2 Tiny |
| Date       | 14 July 2026     | Macro F1 | 0.638            |
| Dataset    | 282 test samples | Accuracy | 73.0%            |

## Predicted Grade

Normal — Grade 0

Confidence: **49.5%**

No significant foraminal narrowing detected. Perineural fat is preserved and the nerve root appears uncompressed.

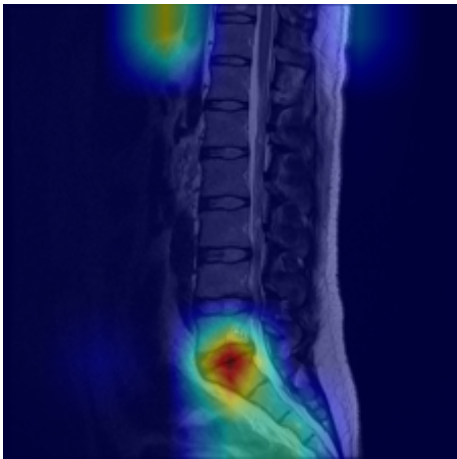
## Class Probability Distribution

| Grade   | Class              | Probability | Confidence                               |
|---------|--------------------|-------------|------------------------------------------|
| Grade 0 | Normal (Grade 0)   | 49.5%       | ████████████████████████████████████████ |
| Grade 1 | Mild (Grade 1)     | 48.2%       | ████████████████████████████████████████ |
| Grade 2 | Moderate (Grade 2) | 0.4%        | ████████████████████████████████████████ |
| Grade 3 | Severe (Grade 3)   | 1.8%        | ████████████████████████████████████████ |

## MRI Images



Original MRI crop (224×224)



Grad-CAM — warmer = higher activation

## Grade Scale Reference

| Grade   | Severity | Clinical Description                                                         |
|---------|----------|------------------------------------------------------------------------------|
| Grade 0 | Normal   | No foraminal narrowing. Perineural fat intact. No nerve root compression.    |
| Grade 1 | Mild     | Mild narrowing. Partial fat obliteration. Nerve root not yet deformed.       |
| Grade 2 | Moderate | Moderate compression. Complete fat obliteration. Nerve root contact present. |
| Grade 3 | Severe   | Severe compression. Nerve root deformation. Surgical consult may be needed.  |

*DISCLAIMER: This report is generated by an AI research tool for educational and research purposes only. It does not constitute a medical diagnosis or substitute for evaluation by a qualified radiologist or physician.*